

EXPERIMENT 10

DATA VISUALIZATION FOR INVENTORY MANAGEMENT SYSTEM

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AIM:

To create data visualizations such as pie charts and bar graphs for an inventory management system to represent stock distribution effectively.

PROCEDURE:

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Inventory Management Visualization</title>
<style>
  body {
    font-family: Arial, sans-serif;
    text-align: center;
    margin: 50px;
  }
  canvas {
    margin: 20px auto;
  }
```

```

    </style>
</head>
<body>
    <h1>Inventory Management System</h1>
    <!-- Pie Chart -->
    <canvas id="pieChart" width="400" height="400"></canvas>
    <!-- Bar Chart -->
    <canvas id="barChart" width="400" height="400"></canvas>
    <!-- Chart.js CDN -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <!-- JavaScript File -->
    <script src="script.js"></script>
</body>
</html>

```

JAVASCRIPT:

```

// Inventory Data
const inventoryData = {
  labels: [
    'Electronics',
    'Clothing',
    'Home Appliances',
    'Books',
    'Toys'
  ],
  datasets: [
    {

```

```

        label: 'Items in Stock',
        data: [200, 150, 100, 80, 50],
        backgroundColor: [
            '#FF6384',
            '#36A2EB',
            '#FFCE56',
            '#4BC0C0',
            '#9966FF'
        ]
    }
]
};

// ----- PIE CHART -----

const ctxPie = document
    .getElementById('pieChart')
    .getContext('2d');

const pieChart = new Chart(ctxPie, {
    type: 'pie',
    data: inventoryData,
    options: {
        responsive: true,
        plugins: {
            title: {
                display: true,
                text: 'Inventory Distribution'
            }
        }
    }
});

```

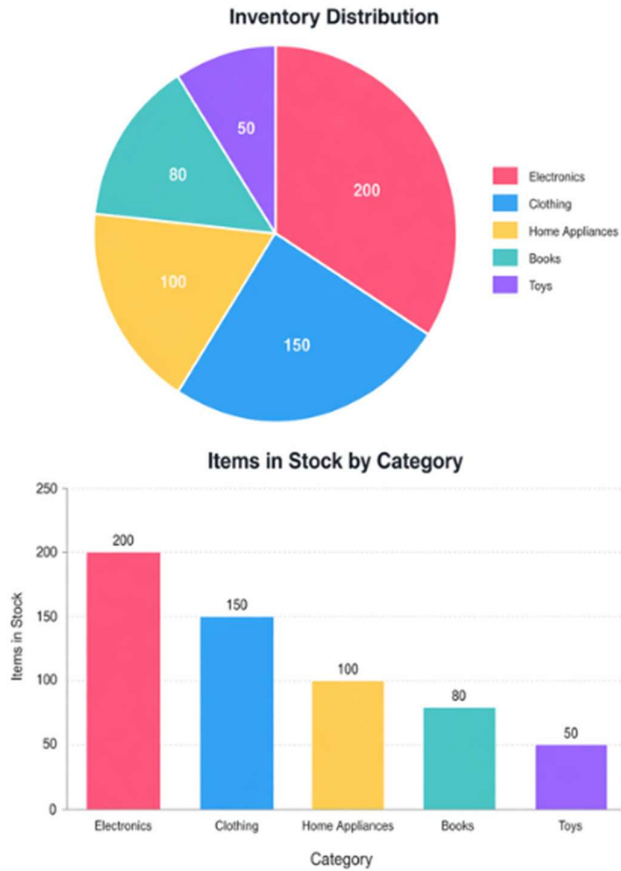
```

    }
  }
});
// ----- BAR CHART -----
const ctxBar = document
  .getElementById('barChart')
  .getContext('2d');
const barChart = new Chart(ctxBar, {
  type: 'bar',
  data: inventoryData,
  options: {
    responsive: true,
    plugins: {
      title: {
        display: true,
        text: 'Items in Stock by Category'
      }
    },
    scales: {
      y: {
        beginAtZero: true
      }
    }
  }
});

```

OUTPUT:

Inventory Management System



RESULT

The pie chart and bar graph successfully represented inventory stock distribution and category-wise stock analysis.